

JOSEPH TAN

Senior Full-Stack Engineer | Technical Lead

Shanghai, China
josephstan607@gmail.com

Professional Summary

Full-Stack developer mostly working on internal business systems, SaaS platforms, and operational dashboards. Most projects involved React, Node.js, PostgreSQL, Redis, and AWS deployment work. Ended up spending a lot of time on frontend performance issues, database bottlenecks, and operational edge cases that only show up after systems are already in production. Prefer practical systems that are easy to maintain over overly abstract architectures.

Core Technical Skills

Frontend: React.js, Next.js, Vue.js, TypeScript, Redux Toolkit, Tailwind CSS, TanStack Table

Backend: Node.js, Express.js, NestJS, Python, Django, GraphQL, event-driven workflows

Databases: PostgreSQL, MySQL, MongoDB, Redis, query optimization

Infrastructure: AWS EC2, Docker, Nginx, Linux, GitHub Actions, CI/CD, deployment automation

UI / UX: Responsive Design, Dashboard UX, Design Systems, User Flow Optimization

Leadership: Agile/Scrum, Sprint Planning, Code Review, Mentoring

Selected Engineering Achievements

- Worked on a warehouse management platform used across 7 regional warehouses. One recurring issue was inventory mismatch during concurrent updates, especially when warehouse staff refreshed pages repeatedly during busy hours. A surprising amount of time went into tracking duplicate writes, delayed sync events, and race-condition-like behavior between services.
- Reorganized several operational dashboards and approval flows. Some of the older screens looked technically correct but confused almost every new staff member during onboarding.
- Built reporting and export tools for operations teams that were previously copying data between Excel sheets every Friday afternoon for reasons nobody could fully explain anymore.
- Spent a lot of time optimizing slow PostgreSQL queries and Redis cache behavior during peak traffic periods. Most improvements came from restructuring query patterns, reducing lock contention, and separating heavier reporting operations from transactional workloads.
- Helped improve frontend performance by reducing unnecessary re-renders, splitting heavier bundles, and cleaning up older polling logic that had quietly grown over time.
- Participated in an ERP migration project where deployment had to be handled gradually because some warehouse teams were still depending on older workflows nobody wanted to touch too

quickly. Part of the migration involved keeping old and new services running together without breaking downstream processes.

Professional Experience

Freelance Senior Full-Stack Developer

Shanghai, China | May 2023 – Present

- Worked with a 4-person freelance team building SaaS and logistics systems for overseas clients.
- Developed backend APIs and deployment infrastructure for logistics systems. Traffic patterns were usually calm until warehouse cutoff hours, then suddenly unpleasant. Some API behavior also had to be adjusted to avoid cascading failures during bulk imports.
- Updated management dashboards after observing how operations staff were using the system day-to-day, especially around repeated approval and inventory actions.
- Fixed several frontend performance issues caused by oversized table rendering and repeated API polling. Large management tables tend to become everyone's problem eventually.
- Built role and permission management features where some access rules ended up being separated by warehouse region because operational responsibilities were never fully standardized. Spent extra time preventing permission inheritance issues after several uncomfortable internal incidents.
- Usually handled deployment preparation, rollback planning, and production issue follow-ups during release periods. Production problems were almost always more interesting than the original feature work.

Full-Stack Developer — QPSoftware

Shanghai, China | Jul 2020 – Mar 2023

Website: <https://www.qpsoftware.com.cn>

- Developed CRM and internal management systems mainly used by manufacturing and retail companies with different operational processes.
 - Helped migrate older jQuery pages into React.js gradually instead of rewriting entire systems at once. Full rewrites sounded good in meetings but rarely made sense in production.
 - Simplified dashboard navigation after repeated feedback from internal users who struggled with older menu structures.
 - Reduced several slow backend endpoints by rewriting heavy joins, adjusting indexing strategies, and moving some processing into async jobs. A few reporting endpoints were indirectly scanning far more data than anyone originally realized.
 - Worked closely with QA during release periods because several client environments behaved differently from staging systems.
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Education

Bachelor's Degree in Computer Science

Xuzhou University of Technology — China

Sep 2016 – Jun 2020

Languages

- Chinese — Native
 - English — Professional Working Proficiency
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Additional Information

- Usually more interested in debugging production behavior and performance bottlenecks than building greenfield demo projects.
- Prefer systems that stay maintainable after 2 years instead of architectures that only look clean in diagrams.
- Spent a lot of time working on operational dashboards, warehouse flows, reporting systems, and frontend-heavy internal tools.
- Comfortable handling both development work and production-side troubleshooting when releases become messy.
- Interested in database behavior, distributed synchronization problems, deployment reliability, and backend system stability under operational load.
- Tend to care quite a bit about permission boundaries, failure handling, and reducing fragile dependencies between services.